

UNIT I

1. Define Irrigation Channel

An **irrigation channel** is an artificial waterway constructed to convey water from a source (like a river, reservoir, or canal) to agricultural fields for the purpose of irrigation. It ensures controlled and efficient water distribution across farmlands.

2. Classify the Channels

Irrigation channels can be classified based on several factors:

- **Based on Lining:**
 - *Lined Channels* (concrete, brick, etc.)
 - *Unlined Channels* (earthen)
 - **Based on Function:**
 - *Main Canal*
 - *Branch Canal*
 - *Distributary*
 - *Field Channel*
 - **Based on Location:**
 - *Contour Channels* (follow land contour)
 - *Side Slope Channels* (run along slopes)
-

3. State Kennedy's Theory

Kennedy's theory states that a channel will be in a state of **critical equilibrium** (i.e., it won't silt or scour) if it flows with a certain **critical velocity (V_c)** which depends on the **depth of flow** and **nature of bed material**.

The critical velocity is given by:

$$V_c = 0.55 \cdot m \cdot D^{0.64}$$

Where,

- V_c = critical velocity (m/s)
 - D = depth of flow (m)
 - m = critical velocity ratio (empirical)
-

4. State Lacey's Theory

Lacey's theory postulates that an alluvial channel achieves **true regime** (stable condition) when it has:

- Uniform flow
- Uniform silt grade and concentration

- Constant discharge
Lacey provided empirical relationships among **velocity**, **hydraulic radius**, **slope**, and **silt factor** to design regime channels.
-

5. Explain Broadly Land Grading

Land grading is the process of reshaping the land surface to a desired **uniform slope** to facilitate uniform water distribution, prevent waterlogging and improve field drainage. It ensures optimum water use efficiency and enhances agricultural productivity.

6. Explain in Detail about Land Leveling

Land leveling is a precise form of land grading. It involves:

- Surveying the field to determine the existing elevations
- Calculating the cut and fill required
- Using equipment like laser-guided levellers or scrapers
- Achieving a uniform slope (or flatness) within a few centimeters of tolerance

Benefits:

- Uniform irrigation
 - Reduced runoff and erosion
 - Increased crop yield
 - Better seed germination
 - Efficient nutrient and water usage
-

7. What are the Design Methods Available to Design the Channel?

The main design methods for irrigation channels include:

1. **Kennedy's Method** – Empirical method based on critical velocity concept.
 2. **Lacey's Method** – Regime theory based on silt factor and discharge.
 3. **Tractive Force Method** – Based on the resistance of channel bed material against erosion.
 4. **Manning's Formula** – Used for calculating velocity and flow in open channels.
 5. **Empirical/Trial and Error Methods** – Using design charts or spreadsheets.
 6. **Computer-Aided Design** – Using software like AutoCAD Civil 3D, SWAT, or HEC-RAS.
-

8. What is the Purpose of Studying Agricultural Water Management in Various Domains?

The purpose includes:

- **Efficient Water Use:** Maximize crop yield with minimal water usage.
- **Sustainability:** Prevent overuse of water resources and maintain long-term productivity.

- **Flood and Drought Mitigation:** Manage excess or deficient water conditions effectively.
- **Irrigation Planning and Design:** Improve irrigation system efficiency.
- **Soil Conservation:** Prevent erosion and degradation.
- **Climate Resilience:** Adapt farming practices to climate variability.
- **Integrated Water Resources Management (IWRM):** Coordinate land and water management at watershed level.

UNIT II

1. What is meant by Command Area?

Command Area refers to the total land area that can be irrigated by a particular irrigation system or canal under normal operating conditions. It is the region that receives water supply from a source like a dam or a canal system.

2. Explain CADA Programme Installation in Tamil Nadu

CADA (Command Area Development Authority) was introduced to improve water use efficiency and ensure better on-farm water management in command areas.

In Tamil Nadu, the CADA programme was implemented under the Command Area Development Programme (CADP) in major irrigation projects like:

- Cauvery Delta
- Parambikulam-Aliyar Project (PAP)
- Amaravathi and Bhavani schemes

Objectives in Tamil Nadu:

- Construction of field channels for uniform water distribution
- Land leveling and shaping to improve productivity
- Introduction of rotational water supply (warabandi)
- Participatory irrigation management through Water Users Associations
- Integrated use of water, land, and crops

3. State Duty of Water

Duty of water is defined as the **area of land (in hectares or acres)** that can be irrigated by a unit discharge of water (usually 1 cusec) flowing continuously during the base period of a crop.

$$\text{Duty (D)} = \frac{\text{Area (A)}}{\text{Discharge (Q)}}$$

4. Define Delta of Water

Delta (Δ) is the total depth of water (in cm or meters) required by a crop during its entire base period from the time of sowing to harvesting.

5. Write the Relationship Between Duty and Delta of Water

5. Write the Relationship Between Duty and Delta of Water

The relationship between Duty (D), Delta (Δ), and Base Period (B) is given by:

$$\Delta = \frac{8.64 \times B}{D}$$

Where,

- Δ = Delta in meters
 - B = Base period in days
 - D = Duty in hectares per cumec (ha/cumec)
 - 8.64 is a conversion factor
-

6. How Do You Explain Rotational Water Distribution?

Rotational water distribution, also called Warabandi, is a method where irrigation water is supplied to farmers in turns at fixed time intervals and for fixed durations.

This ensures:

- Equity in water distribution
 - Predictable scheduling
 - Efficient utilization and conflict minimization among farmers
-

7. Classify the Command Area

Command Area can be classified as:

- Gross Command Area (GCA):
Total area that can be irrigated by an irrigation project, including cultivable and uncultivable land.
 - Culturable Command Area (CCA):
Portion of GCA that is arable and suitable for cultivation.
 - Irrigated Command Area (ICA):
Portion of CCA that is actually irrigated in a specific season or year.
-

8. Explain the Term of Irrigation

Irrigation is the artificial application of water to the soil to assist in the growth of crops, especially when rainfall is insufficient.

It is crucial for:

- Ensuring crop survival and yield
- Maintaining soil moisture
- Enabling multiple cropping
- Supporting agricultural productivity in dry or semi-arid areas

Types of Irrigation:

- Surface Irrigation (Furrow, Basin, Border)
- Sprinkler Irrigation
- Drip Irrigation
- Subsurface Irrigation

UNIT III

1. Compare and Contrast Surface and Sub-surface Water

Aspect	Surface Water	Sub-surface Water (Groundwater)
Definition	Water found on the surface of the Earth	Water located beneath the Earth's surface in soil/rocks
Sources	Rivers, lakes, ponds, reservoirs	Aquifers, wells, infiltration galleries
Accessibility	Easily accessible for direct use	Needs pumping or bore wells to extract
Contamination Risk	Higher (due to pollution, runoff)	Lower (natural filtration through soil)
Usage	Common in surface irrigation and hydro projects	Used for drinking, domestic use, and tube well irrigation

2. Tabulate the Availability of Water in the World

Water Source	Percentage of Total Water
Oceans (Saline)	97.5%
Freshwater	2.5%
– Glaciers and Ice Caps	68.7% of freshwater
– Groundwater	30.1% of freshwater
– Surface Water (rivers/lakes)	1.2% of freshwater

3. Explain the Term of Ground Water

Groundwater is the water that fills the pores and fractures in underground soil and rock layers. It originates from precipitation that infiltrates the ground and gets stored in aquifers.

It serves as a vital resource for:

- Drinking water
 - Irrigation
 - Industrial and domestic uses
-

4. List the Various Requirements of Water in Different Crops

Crop	Water Requirement (mm/season)
Paddy (Rice)	1200 – 1500
Wheat	450 – 650
Sugarcane	1500 – 2500
Cotton	700 – 1200
Groundnut	500 – 600
Maize	500 – 800
Vegetables	400 – 600

5. State the Term of Canal Supply

Canal supply refers to the regulated release and distribution of water through a canal network from a storage structure (like a dam or reservoir) to the command area for irrigation purposes. It is usually controlled by irrigation departments and supplied according to crop demand and schedule.

6. What is Meant by Irrigation Demand?

Irrigation demand is the amount of water required to supplement the natural rainfall for a particular crop to meet its optimal growth and yield.

It depends on:

- Crop type and growth stage
 - Soil characteristics
 - Climate conditions
 - Irrigation method used
-

7. What Are the Methods Available to Predict the Rainfall?

Rainfall prediction methods include:

1. Synoptic Methods (weather forecasting based on atmospheric parameters)
 2. Empirical or Statistical Models (based on past rainfall data trends)
 3. Numerical Weather Prediction (NWP) – Uses mathematical models to simulate weather
 4. Remote Sensing and Satellite Imaging
 5. Weather Radar Observations
 6. Artificial Intelligence & Machine Learning – For short-term and long-term forecasting
-

8. What is Meant by Rainfall? Explain Major Months of Rainfall in Tamil Nadu.

Rainfall refers to precipitation in the form of water droplets from the atmosphere to the Earth's surface due to condensation of moisture in clouds.

Major Months of Rainfall in Tamil Nadu:

- Southwest Monsoon: June to September (less dominant in TN)
- Northeast Monsoon: October to December (Primary rainy season for Tamil Nadu)
 - Around 48% of annual rainfall occurs during this period
- Summer Showers: April–May (isolated convective rains)